

JAYESH DONGRE

XR & Spatial AI Engineer | Real-Time 3D, Computer Vision & Gaussian Splatting

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

XR and spatial AI engineer with 4+ years building real-time 3D systems and applied-AI experiences across healthcare and defense. I work mainly in point cloud streaming, RGB-D 3D reconstruction, and multiplayer XR on Unity, Unreal Engine, and Apple Vision Pro (visionOS), and I pair that with computer vision and deep learning across medical-image segmentation, Gaussian splatting, on-device LLMs, and RAG-based speech interfaces. I've shipped multi-user immersive applications on Meta Quest, visionOS, iOS, Android, and Windows, backed by low-latency streaming on Cloudflare and AWS.

CORE SKILLS

Engines & XR: Unity (C#), Unreal Engine (C++, Blueprints), OpenXR, Apple Vision Pro / visionOS, RealityKit, Meta Quest, ARKit, ARCore, AR Foundation, Pixel Streaming, Authoritative Multiplayer Systems

Computer Vision & Real-Time 3D: Computer Vision, Deep Learning, YOLO / Object Detection, Semantic Segmentation, 3D Reconstruction, Point Cloud Streaming, RGB-D Synchronization, Camera Intrinsics & Calibration, Open3D / PCL, OpenCV, NeRF / Gaussian Splatting, Shaders (HLSL)

AI / ML: Deep Learning, Computer Vision, On-Device LLM Integration, Retrieval-Augmented Generation (RAG), Multimodal Interaction (Speech-to-Text / Text-to-Speech), PyTorch

Languages: C#, C++, Python, Swift

EXPERIENCE

XR Engineer | Ethosh Design Pvt Ltd

Pune, Maharashtra, India

Apr 2025 - Present

- Built a real-time VR experience that captures surgical scenes with a ZED depth camera, splits each point-cloud frame into RGB and depth maps along with the camera intrinsics and metadata, and streams that over the internet via WebRTC, so the point cloud is rebuilt on remote VR devices for live, synchronized multi-user viewing.
- Created a casting-style service that lets anyone watch what's happening inside a live VR session straight from a browser tab, and used Cloudflare (Workers / Stream / R2) to scale it out to many simultaneous viewers.
- Developed a VR experience for a neuro-rehabilitation device that ingests live biofeedback data from the device over BLE, visualizes it through a custom immersive experience, and saves each session to a web dashboard and an AWS S3 bucket.
- Integrated an offline, on-device LLM into an Apple Vision Pro (visionOS) application so a user can ask natural-language questions about any product and get answers back, with retrieval-augmented generation (RAG) and speech-to-text for voice input. All inference runs locally, so no data leaves the device.
- Built a Unity app and Gaussian splatting workflow for turning captures into viewable 3D: photos or video are processed with FastGS to generate the Gaussian splat, and once it's ready Unity picks it up and plays it back like a video player, showing the right splat at the right point in the timeline.
- Wrote a reusable Unity plugin that uploads images and files to AWS S3, now used as a drop-in component across several internal projects.

Associate XR Developer | Ethosh Design Pvt Ltd

Pune, Maharashtra, India

Aug 2022 - Mar 2024

- Developed a VR rifle training simulator replicating an exact firearm and its real-world gun mechanics, validated and approved by Army majors, reducing reliance on live-fire range time.
- Built a multiplayer VR knee-surgery simulation in Unreal Engine where multiple surgeons can join a shared session to interact, perform the procedure together, and ask questions in real time, using authoritative server replication.
- Created a CT-scan segmentation and 3D reconstruction pipeline (computer vision / deep learning) that takes a knee CT scan as input and outputs segmented anatomy and a reconstructed 3D model at clinical-grade sub-millimeter (~2 mm) accuracy.

- Delivered a cross-platform XR product-visualization app (iOS, Android, Windows) for medical-device training and clinician / sales enablement.
- Optimized rendering pipelines on Meta Quest and mobile to hold a stable 72 FPS in complex scenes via draw-call batching, LODs, and occlusion culling, cutting frame time by about 30%.

XR Developer Intern | Ethosh Design Pvt Ltd

Feb 2022 – Jul 2022

Pune, Maharashtra, India

- Built Unreal Engine Pixel Streaming applications enabling browser-based remote access to high-fidelity 3D environments.
- Integrated MetaHuman interactive agents, increasing user engagement and demo realism in simulation environments.
- Authored modular, reusable components that shortened feature-development cycles across the team.

SELECTED PROJECTS

FastGS: 3D Gaussian Splatting & Mesh Generation [GitHub](#)

- End-to-end 3DGS pipeline that trains a Gaussian Splatting scene from raw images or video and extracts a textured mesh (PLY). Built on FastGS / PGSR with a COLMAP front-end, a custom CUDA rasterizer patch, depth-normal and multi-view NCC geometry losses, and TSDF-fused mesh export. Stack: Python, CUDA, C++, PyTorch.

Knee Implant Pipeline: CT-to-Mesh for Surgical Planning [GitHub](#)

- Automated clinical pipeline that turns a raw knee CT scan (DICOM) into a clean, validated 3D bone mesh (STL) for orthopedic surgery planning. It preprocesses the scan (HU windowing, denoising, isotropic resampling), runs AI segmentation with TotalSegmentator v2 to isolate the femur and tibia, then builds and refines the surface mesh with Marching Cubes, Taubin smoothing, and QEM decimation. It can also measure medullary canal centerlines and diameters, and validates each result against an HU-threshold baseline using Dice and Hausdorff metrics, with review in 3D Slicer. Stack: Python, PyTorch, MONAI, TotalSegmentator v2, PyMeshLab, SimpleITK, scikit-image.

Unity WebRTC Video Streaming [GitHub](#)

- Low-latency video-streaming system in Unity over WebRTC, with a WebSocket signaling server and STUN / TURN NAT traversal for peer-to-peer streaming between Unity and Python (aiortc) clients. Stack: C#, Unity WebRTC, WebSockets.
- **Note:** Applied this same WebRTC approach to stream real-time point clouds over Cloudflare in a production system; that implementation is under NDA and cannot be shared publicly.

EDUCATION

Master of Computer Applications (MCA), Fergusson College, Pune

2022

B.Sc. Computer Science, Fergusson College, Pune

2020